Institute of Geographical Information Systems
National University of Sciences & Technology

Lab Description:

Researchers quote NDVI products have been shown to be unstable, varying with soil color, soil moisture, and saturation effects from high density vegetation. In an attempt to improve NDVI, Huete developed a vegetation index that accounted for the differential red and near-infrared extinction through the vegetation canopy. The index is a transformation technique that minimizes soil brightness influences from spectral vegetation indices involving red and near-infrared (NIR) wavelengths.

The index is given as:

$$\mathrm{SAVI} = \frac{(1 + \mathrm{L})(\mathrm{NIR} - \mathrm{Red})}{(\mathrm{NIR} + \mathrm{Red} + \mathrm{L})}$$

Where L is a canopy background adjustment factor. An L value of 0.5 in reflectance space was found to minimize soil brightness variations and eliminate the need for additional calibration for different soils. The transformation was found to nearly eliminate soil-induced variations in vegetation indices**.**

Your task is to calculate SAVI & NDVI using the provided imagery.

Data Set:

tarbela 150036.img

Tools for Project:

- NDVI Calculation
- Reclassification
- Calculation of SAVI using Erdas Model Builder
- Extraction of suitable classes using SAVI
- Comparison of SAVI & NDVI (Image, Classes)
- Generate Maps for SAVI & NDVI

Required lab report contents:

- Explain all steps performed with screen shot where appropriate
- Demonstrate the final output to Instructor.